

Speaking Script — Slides 9 and 10

Adaptive thresholds, and the audit of Fairlearn and AIF360. Everything on each slide, every chart explained, why each result happened, the mathematics used, and the questions your professor is most likely to ask. Normal text is what you say out loud. Green is background you should understand but need not read aloud. Red is a warning or a mistake we made. Speaking time: about 7 minutes for both slides.

SLIDE 9 — We let the model choose the bar itself

What is on this slide

Where	What it is
Top	Title, and a subtitle crediting the idea to you, Sir
Top-left chart	Two panels. Left: the threshold the model picked per group. Right: the share of real crimes caught in each group
Caption	One line explaining what Arm A was told to do
Bottom-left chart	Three bars comparing the gap under each method
Card 1 (pink)	Arm A — actively harmful
Card 2 (green)	Arm B — honest, but not a fix
Card 3 (grey)	The giveaway: AUC barely moved

The opening — give him the credit first

"Sir, you suggested we try adaptive thresholds. Not a fixed rule like 0.5 for everybody, and not a formula we write by hand, but a bar the model works out for itself from the data. We built that, and the result surprised us."

What a threshold actually is — say this if he wants the basics

"The model does not output yes or no. It outputs a number between 0 and 1, like a weather forecast: 0.7 means seventy percent chance of crime here this week. To act, somebody has to pick a cut-off. Above the cut-off we send patrols; below it we do not. Normally that cut-off is 0.5, chosen by habit, not by evidence. The question was whether letting the model choose it per neighbourhood group would close our gap."

The two versions we built

Arm	What we told it to do	Result
A	Pick the bar in each group that gives the best F1 score in that group	Opened a 35.56 point gap in crimes caught
B	Pick the bar in each group so the false-alarm rate is the same everywhere	Gap of -1.57 — behaved correctly

Explaining the left chart

"Look at the left panel. Arm A chose 0.425 for busy areas, 0.520 for the middle, and 0.601 for poor areas. It set the highest bar for the poorest neighbourhoods. Now the right panel shows what that did. In busy areas it caught 89.1 percent of the crimes that really happened. In poor areas, 53.5 percent. A gap of 35.6 points."

"And here is the part that matters. We built this data ourselves, so we know the truth. Every group was equally easy to predict — identical by construction. There was no real difference in how well the model understood rich and poor areas. The gap you are looking at was created entirely by the fix."

Why did it choose a higher bar for poor areas? — be ready for this

"Because the model was trained on all areas together, its output numbers are tuned to the average crime rate across the whole city. In a poor, low-crime area those numbers come out too high — the model is over-confident there. So to get a sensible answer the bar has to be raised. That is arithmetic, not prejudice. But the effect on people is the same: fewer real crimes in poor areas get flagged."

Background: this is why our result appears to differ from Lipton et al. (2014), who showed the F1-optimal threshold equals half the optimal F1 for *calibrated* probabilities. Our scores are calibrated to the pooled base rate, not per group. If he raises this, that is the answer — and it is a point in our favour, because it shows we know the relevant literature.

Explaining the bottom chart

"The bottom chart compares all three. One bar for everyone gives a gap of minus 0.13 — essentially zero, which is correct, because the groups really are equal. Arm A gives plus 35.56. Arm B gives minus 1.57. Same model, same data, same skill. Only the bar moved."

The giveaway — this is the sentence to land

"And here is how we know nothing real changed. AUC — the measure that does not depend on where you put the bar — moved by 0.07 in all three cases. Seven hundredths of a point. The model never improved and never got worse. It was the same model throughout. Everything you see in that chart is the bar moving, and the score following it."

The conclusion to state plainly

"So adaptive thresholds are not a fix. Arm A is worse than doing nothing — it manufactures the exact harm we were trying to remove. Arm B is honest but adds no skill; it only stops us fooling ourselves. What this experiment gave us is not a solution. It is a measuring instrument. It shows how far the reported score can move while the model stands still."

The mathematics — if he asks how we optimised a threshold

A threshold is a step function, so it has zero gradient and cannot be trained directly. We replaced the hard comparison with a sigmoid, which is smooth and therefore differentiable:

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p_i = sigmoid( beta * ( s_i - tau ) ) with beta = 50
TP = sum_i p_i * y_i  FP = sum_i p_i * (1 - y_i)
FN = sum_i (1 - p_i) * y_i  TN = sum_i (1 - p_i) * (1 - y_i)
soft-F1 = 2*TP / ( 2*TP + FP + FN )
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"With beta = 50 the sigmoid is almost a step, so the soft count is very close to the true count, but it still has a gradient. We then optimise tau for each group by ordinary gradient descent. Arm A maximises soft-F1. Arm B minimises the squared difference between each group's false-alarm rate and the pooled rate at 0.50."

A mistake we made, and caught — worth telling him

Our first version of Arm B minimised the spread in *true*-positive rate rather than false-positive rate. That objective is degenerate: it drove every threshold down to 0.02, so the model predicted crime everywhere and every group reached 100 percent of crimes caught — a perfect score for a useless model. We found it in an independent NumPy check before trusting any output, and rewrote the objective to match false-alarm rates instead. Mentioning this shows you validate your own work.

Honest limitation to state before he finds it

"One limitation, Sir. These threshold numbers are from synthetic data, where we could guarantee the groups were equally easy to predict. That is what makes the result clean — we know the truth. We have not yet run this on the real Chicago scores. That is on the list."

SLIDE 10 — We tested the fairness tools everyone uses

What is on this slide

Where	What it is
Top	Title and subtitle naming both libraries
Top-left chart	Five horizontal bars: the disparity each metric reported. Red = wrong, green = right
Caption	How the data was built, and what the honest answer was
Bottom strip, left	Fairlearn's default: the gap before and after the 'fix'
Bottom strip, right	AIF360: wrongly-flagged crime-free places in poor areas, before and after
Card 1 (pink)	The default setting is the bad one
Card 2 (pink)	IBM's tool flagged 3x more
Card 3 (green)	Verified: same numbers on Kaggle and locally

The opening

"So far everything I have shown you is about our own model. The obvious question is whether this matters outside our project. So we tested the two fairness libraries that thousands of people actually use in industry — Fairlearn, from Microsoft, and AIF360, from IBM."

Explain the test setup carefully — the whole slide depends on it

"We built a dataset where we controlled the truth. Three groups. In every group, the relationship between the data and the crime is exactly the same — identical. So all three groups have the same ROC curve and the same AUC. The model is equally good at all of them. The only thing we changed is how much crime there is: 60 percent, 35 percent, 12 percent."

"On this data the correct audit result is: there is nothing to fix. Any tool that reports a big problem is reacting to the amount of crime, not to unfairness."

Technically: $\text{score} | y=1 \sim N(1,1)$ and $\text{score} | y=0 \sim N(0,1)$, identical in all three groups; base rates 0.60 / 0.35 / 0.12. We verified equal skill by hand before using any library — at a matched false-positive rate of 5 percent the three groups caught 30.2, 25.9 and 29.9 percent of real events. Equal, within sampling noise.

Explaining the top chart

"Here is what Fairlearn's own measurement tool reported. Precision: a 52.67 point disparity. F1: 23.52 points. Accuracy: 18.06 points. Those are the numbers a practitioner would see, and they say there is a serious fairness problem."

"Now look at the bottom two bars. Recall reported 2.18. AUC reported 0.84. Those two are correct. The groups genuinely are equal, and only those two metrics can see it."

Explaining the bottom-left chart — the strongest point

"That was the measurement side. This is the repair side, and it is worse. Fairlearn has a tool that adjusts thresholds per group to remove unfairness. Its default setting is called demographic parity, which means: flag every group equally often."

"Before the fix, the gap in crimes caught was plus 0.35 — correct, no gap. After the fix, minus 21.26. It opened a 21 point gap where there was none. We ran it five times; the standard deviation is 0.52, so this is not noise."

"And I want to stress the word default. This is what you get if you install the library and do not change any setting. The harmful option is the one a non-expert receives for making no choice at all."

The per-group numbers, if he wants detail

Group	Flagged	Crimes caught	False alarms
Busy	24.2%	35.50%	7.12%
Middle	24.3%	46.07%	13.05%
Poor	24.3%	56.33%	19.84%

"Everybody is flagged equally often — about 24 percent — which is exactly what the tool promised. But poor areas now have a false-alarm rate of 19.84 percent against 7.12 for busy areas. Nearly three times as many crime-free places wrongly flagged, in the group that can least afford it."

Explaining the bottom-right chart

"IBM's library does the same thing more severely. Its Reject Option method, set to statistical parity, took wrongly-flagged crime-free places in poor areas from 13.49 percent to 41.55 percent. Three times more. In a policing system, those are patrols sent to places where nothing was going to happen."

"And again, AUC did not move. 76.9 for busy areas, 76.0 for poor areas, on every single row. The model was never touched. Only the thresholds moved."

Be fair to the libraries — say this, it protects you

"I should be fair to these tools. The settings that target crimes-caught directly — equal opportunity, and equalized odds — behaved correctly in our test, with gaps near zero. So this is not a blanket accusation. The failure is specific: it is the family of settings that equalise how often each group is flagged, and that family happens to contain the default."

The honesty point — do not skip this

The conflict between demographic parity and equal opportunity when base rates differ is a **known theoretical result** — Kleinberg, Mullainathan and Raghavan (2016), and Chouldechova (2017). We must not present it as our discovery. A reviewer would reject the paper on that alone.

"Sir, I want to be clear about what is ours and what is not. That these two fairness definitions conflict is known theory from 2016 and 2017, and we cite it. What we are adding is narrower: we show that this known conflict is the default setting of widely used software, and we put a number on what it costs — a 21 point gap, and three times the wrongful flagging. That is a smaller claim, but it is defensible."

Questions he is likely to ask

Question	Your answer
Is synthetic data fair to these tools?	It is the only way to know the ground truth. On real data we could never be certain the groups are equally predictable, so we could not tell a correct alarm from a false one. The synthetic case is the clean test; real data is the next step.
Are you saying nobody should use Fairlearn?	No. We are saying the default setting should not be used without thought, and that AUC should be reported alongside whatever fairness metric is chosen.
Did you verify this independently?	Yes. Same numbers in two environments, five seeds, and the equal-skill assumption checked by hand in NumPy before any library was called.
What should people do instead?	Report a threshold-free metric such as AUC next to the fairness metric. If the fairness number moves and AUC does not, nothing about the model has changed.

How to close both slides

"Taken together with the earlier results, this is now the same finding shown four separate ways: in our graph models, in the threshold sweep, in adaptive thresholds, and in the two libraries the industry uses. The score everyone reports moves a long way without the model getting any better. That is the paper."